

MT 1003 – Calculus and Analytical Geometry

Homework #1

Domain, Range, and Graphing of Functions

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Instructions:

1. All homework files must follow the naming format

Section_RollNumber_HW1

(for example, A.26I1234_HW1). Submissions that do not follow this format will not be evaluated.

2. Write your name and roll number clearly on **every page**.
3. Submit your complete homework as a **single scanned PDF file**.
4. For Section B, include the Python code, generated graphs, and your written observations in the same PDF file.

Failure to comply with these instructions may result in zero marks.

Section A: Analytical Work

Q1. Memory Usage. A program's memory usage is modeled by

$$M(n) = \sqrt{100 + 20n - n^2},$$

where n represents the number of processes running simultaneously.

- (a) Determine the mathematical domain and range of $M(n)$.
- (b) Find the intercepts.
- (c) Sketch the graph.
- (d) Taking account of the meaning of n , state the practical domain and range and explain them in context.

Q2. Algorithm Performance. The running time of an algorithm is approximated by

$$T(n) = \frac{3n + 2}{n - 4},$$

where n represents the input size. Determine the mathematical domain and range. Then state an appropriate practical restriction on n .

Q3. Machine-Learning Activation Function. Consider the function

$$f(x) = \ln |x - 2| + 1.$$

- (a) Determine its domain and range.
- (b) Find the intercepts.
- (c) Sketch the graph.
- (d) Explain why $x = 2$ is excluded from the domain.

Q4. Network Signal Strength. The strength of a normalized network signal is modeled by

$$S(d) = \frac{10}{\sqrt{d^2 + 4}}, \quad d \geq 0,$$

where d is the distance from the transmitter. Determine the domain and range.

Q5. Data-Compression Error. A simplified compression-error model is

$$E(r) = |r^2 - 4r + 3|, \quad 0 \leq r \leq 4,$$

where r is a compression parameter.

- (a) Determine the domain and range on the given practical interval.
- (b) Find all intercepts.
- (c) Determine where $r^2 - 4r + 3$ is positive and where it is negative.
- (d) Sketch the graph.
- (e) Explain how the absolute value affects the shape of the graph.

Section B: Introduction to Graphing with Python

This section requires **no previous knowledge of Python**. Follow the steps below, run the complete code, and answer the questions by examining the generated graphs.

Installing Anaconda and Opening Jupyter Notebook

1. Download the Anaconda Distribution from www.anaconda.com/download and install it using the recommended options.
2. Open **Anaconda Navigator** from the Start menu.
3. Find **Jupyter Notebook** and click **Launch**.
4. In the browser window, open the folder in which you want to save your work.
5. Select **New** → **Python 3 (ipykernel)**.
6. Copy the code below into one notebook cell. Press **Shift+Enter** to run it.

Q6. Plotting Three Functions. Run the following complete program. The comments beginning with # explain the purpose of each line.

```
1  # Import NumPy for creating numerical input values and
   # calculations.
2  import numpy as np
3
4  # Import pyplot for drawing and displaying graphs.
5  import matplotlib.pyplot as plt
6
7  # Create 1000 x-values from -5 to 15 for the memory function.
8  x1 = np.linspace(-5, 15, 1000)
9
10 # Calculate the expression inside the square root.
11 inside = 100 + 20*x1 - x1**2
12
13 # Keep only the x-values for which the square root is real.
14 valid = inside >= 0
15
16 # Compute the memory function at all valid x-values.
17 y1 = np.sqrt(inside[valid])
18
19 # Create x-values on both sides of x = 4 for the rational
   # function.
20 x2_left = np.linspace(-10, 3.9, 500)
21 x2_right = np.linspace(4.1, 15, 500)
22
23 # Compute the rational function on the left of x = 4.
24 y2_left = (3*x2_left + 2)/(x2_left - 4)
```

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25
26 # Compute the rational function on the right of  $x = 4$ .
27  $y2\_right = (3*x2\_right + 2)/(x2\_right - 4)$ 
28
29 # Create input values from 0 to 4 for the compression-error
    function.
30  $x3 = np.linspace(0, 4, 500)$ 
31
32 # Compute the absolute-value compression-error function.
33  $y3 = np.abs(x3**2 - 4*x3 + 3)$ 
34
35 # Create one figure containing three graphs arranged in one
    row.
36  $fig, axes = plt.subplots(1, 3, figsize=(16, 4.5))$ 
37
38 # Draw the memory-usage graph in the first position.
39  $axes[0].plot(x1[valid], y1, color="blue", linewidth=2)$ 
40
41 # Add a title to the first graph.
42  $axes[0].set\_title(r"\$M(x)=\sqrt{100+20x-x^2}\$")$ 
43
44 # Label the horizontal axis of the first graph.
45  $axes[0].set\_xlabel("x")$ 
46
47 # Label the vertical axis of the first graph.
48  $axes[0].set\_ylabel("M(x)")$ 
49
50 # Draw horizontal and vertical coordinate axes on the first
    graph.
51  $axes[0].axhline(0, color="black", linewidth=0.8)$ 
52  $axes[0].axvline(0, color="black", linewidth=0.8)$ 
53
54 # Add a grid to the first graph.
55  $axes[0].grid(True, alpha=0.3)$ 
56
57 # Draw both branches of the rational function in the second
    position.
58  $axes[1].plot(x2\_left, y2\_left, color="red", linewidth=2)$ 
59  $axes[1].plot(x2\_right, y2\_right, color="red", linewidth=2)$ 
60
61 # Show the excluded input  $x = 4$  with a dashed vertical line.
62  $axes[1].axvline(4, color="black", linestyle="--", linewidth$ 
     $=1)$ 
63
64 # Add the title and axis labels to the second graph.
65  $axes[1].set\_title(r"\$T(x)=\frac{3x+2}{x-4}\$")$ 
66  $axes[1].set\_xlabel("x")$ 

```

```

67 axes[1].set_ylabel("T(x)")
68
69 # Restrict the displayed y-values so the graph is easy to
    read.
70 axes[1].set_ylim(-20, 20)
71
72 # Draw the coordinate axes and grid on the second graph.
73 axes[1].axhline(0, color="black", linewidth=0.8)
74 axes[1].grid(True, alpha=0.3)
75
76 # Draw the compression-error graph in the third position.
77 axes[2].plot(x3, y3, color="purple", linewidth=2)
78
79 # Add the title and axis labels to the third graph.
80 axes[2].set_title(r"$E(x)=|x^2-4x+3|$")
81 axes[2].set_xlabel("x")
82 axes[2].set_ylabel("E(x)")
83
84 # Draw the coordinate axes and grid on the third graph.
85 axes[2].axhline(0, color="black", linewidth=0.8)
86 axes[2].axvline(0, color="black", linewidth=0.8)
87 axes[2].grid(True, alpha=0.3)
88
89 # Adjust the spacing so titles and labels do not overlap.
90 plt.tight_layout()
91
92 # Display all three graphs.
93 plt.show()

```

- Paste a screenshot of the three graphs into your homework.
- From the first graph, estimate the domain and range of $M(x)$. Then verify your answer algebraically.
- From the second graph, identify the excluded x -value and the horizontal value that the graph cannot attain. Use these observations to state the domain and range of $T(x)$.
- From the third graph, state the domain and range of $E(x)$ on $0 \leq x \leq 4$.
- Identify the intercepts visible in each graph.
- In two or three sentences, explain one advantage and one limitation of determining domain and range only from a computer-generated graph.

Q7. Plotting a Regression Line. A simple regression model predicts a student's quiz score y from study time x by

$$y = 4x + 3, \quad 0 \leq x \leq 5.$$

Add a new notebook cell and run the following code.

```
1  # Import NumPy for numerical values.
2  import numpy as np
3
4  # Import pyplot for graphing.
5  import matplotlib.pyplot as plt
6
7  # Create 100 study-time values from 0 to 5 hours.
8  x = np.linspace(0, 5, 100)
9
10 # Use the regression equation to predict the quiz scores.
11 y = 4*x + 3
12
13 # Plot the predicted score against study time.
14 plt.plot(x, y, color="darkgreen", linewidth=2)
15
16 # Add a descriptive title to the graph.
17 plt.title("Regression Model: Predicted Quiz Score")
18
19 # Label the horizontal and vertical axes.
20 plt.xlabel("Study time (hours)")
21 plt.ylabel("Predicted quiz score")
22
23 # Add a grid to make values easier to read.
24 plt.grid(True, alpha=0.3)
25
26 # Display the completed graph.
27 plt.show()
```

- Paste a screenshot of the graph into your homework.
- State the domain and range of the regression function in this context.
- What score does the model predict for $x = 2.5$ hours of study?
- Explain the meanings of the slope 4 and the vertical intercept 3.
- Should this model be used for negative study time or for study time far beyond the given interval? Briefly justify your answer.